Does the length of a stellar outburst help forecast the next?

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A star that flares every few weeks

SS Cygni is a pair of stars in which a white dwarf pulls gas from a companion into a surrounding disc. Every seven weeks or so the disc heats up, and the system brightens about twentyfold for one or two weeks before fading (Figure A). Amateur astronomers have recorded its brightness by eye for well over a century, and the American Association of Variable Star Observers (AAVSO) holds their reports. Long and short outbursts tend to alternate, and earlier work found that longer outbursts tend to be followed by longer quiet spells (Cannizzo & Mattei 1992).

We asked whether knowing how long an outburst lasted improves a forecast of when the next one starts, beyond what the recent quiet spells already tell. Two ideas suggest it could. A long outburst may drain more of the disc, which then takes longer to refill. Or outburst length may track a slowly changing feeding rate that recent quiet spells capture only in part. A test fixed before the data were seen gave the answer: yes, by a small margin.

A forecast test fixed in advance

We built two forecasters from the 1910–1959 observations. Both use the recent quiet spells; one also uses the length of the outburst that has just ended. Whenever the reports show that an outburst has faded, each forecaster spreads probability over the possible start dates of the next one. We froze both, published the protocol online, and only then opened the observations from 1960–2017. A forecast earns more credit the more probability it placed on the start window that the later reports revealed.

Outburst length helped

Over 441 forecasts, the forecaster that knew outburst length gained 0.0165 nats of log score per forecast (90% interval 0.0049 to 0.0281), passing the success rule set beforehand. With recent timing history held fixed, doubling outburst length increased the model's median full quiet spell, measured from the estimated end to the next start, by about 9%. The gain is about a seventh of what knowing the recent quiet spells adds, and it was positive in eight of the ten six-year blocks (Figure B). The two other checks planned in advance, on the 1993–2017 years alone and on a long/short forecasting rule, were inconclusive.

After an external review, we re-timed the outbursts in other ways. Scoring against single estimated start dates gave a similar gain, 0.0154 nats. Start windows rebuilt directly from observers' reports gave smaller gains, 0.0075 and 0.0043 nats, with intervals that include zero.

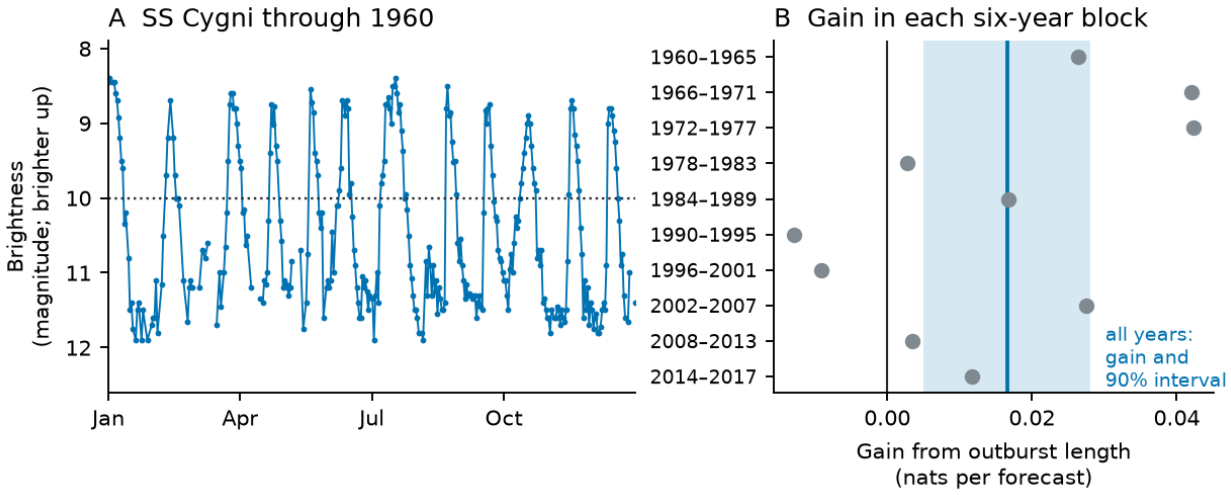


Figure. (A) SS Cygni's nightly brightness through 1960, the first year of the test; the dotted line is the threshold that defines an outburst. (B) The gain from knowing outburst length in each six-year block of 1960–2017; the blue line and band show the overall gain and its 90% interval.

Could other stars be tested?

Before opening their early records, we froze a screen of ten other dwarf novae, the class of star SS Cygni belongs to. None had pre-1990 records that qualified for the same test. The test needs unbroken watching across two successive cycles, and most of these stars are fainter than SS Cygni and were watched less continuously. In the 1980s SS Cygni's record gave 67 usable forecasts for 73 outbursts; the ten others together gave 102 for 485. Their later records, and automated sky surveys, may supply more.

What would tell the two ideas apart

Both ideas fit the result. Forecasts that also include an independent measure of the disc's state, such as its brightness between outbursts or its X-ray output, could separate them: if outburst length stops helping once the state is known, the feeding-rate idea gains support; if it keeps helping, the draining idea does. The frozen forecasters can also be scored on SS Cygni's observations from 2018 onward, which remain unopened.

Limits

- The score rewards probability placed on start windows that software estimated from the reports. Turning it into skill at predicting the true start needs a tested model of when observers look and report.
- An outburst's length and the following quiet spell share one estimated end time, and observers may watch more closely when the star is active. Both could tie outburst length to timing.
- The re-timing checks and baseline recalibrations were designed after the result and reuse the opened data. A richer timing model could absorb part of the gain.
- Historical upload and revision dates are unknown; the test uses the archive as it stands today.
- A documented repair to the data reader followed the first data access, before any forecast was scored.
- The result concerns one star.

How this work was done

Barry Brook set the direction and access rules and decides release. GPT-6 Pro supplied the initial ideas and an external review. Claude Opus 5.5 reviewed the literature, built the outburst finder and drafted the current article and brief. GPT-6 Codex built and ran the forecasts and independently checked the results, the later diagnostics and the drafts. The repository keeps the frozen calculation, every correction and the evidence behind each claim.

Sources

Cannizzo & Mattei (1992) analysed the historical duration–gap relation; Cannizzo (1993) modelled the disc cycle. The protocol is archived at https://doi.org/10.5281/zenodo.24053338.

Cannizzo, J. K. 1993, The Astrophysical Journal, 419, 318
Cannizzo, J. K., & Mattei, J. A. 1992, The Astrophysical Journal, 401, 642